

The D-class interior zeros at $(3, 3, 1, 1, 1)$ are row-zero: a two-sided kernel dichotomy and the anti-involution mirror

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Abstract

At $\lambda = (3, 3, 1, 1, 1)$, where $\widehat{\lambda} = (3, 3)$, $q = 3$ trailing 1s, and $\tau(\widehat{\lambda}) = 0$, the chain operator $\Omega = R'_6 R'_7 R'_8 R'_9$ on V^λ has 108 zero diagonals $p_T(q) = 0$ out of 120 SYTs. The smoking-gun cases of seventeen May ($T_0 = ((1, 2, 5), (3, 4, 6), (7), (8), (9))$ and 25 siblings) were closed by the leftmost-factor SC theorem [1]. We close *all* remaining zeros via a single structural fact:

$$p_T(q) = 0 \iff \Omega v_T = 0 \text{ or } \Omega^* v_T = 0,$$

where $\Omega^* = L'_n L'_{n-1} \cdots L'_{\ell+1}$ with $L'_k = S_1 S_2 \cdots S_{k-1}$ is the anti-involution image of Ω in $H_q(S_n)$. The first kernel (column-zero) reflects right-multiplicative absorption at the bottom of the chain; the second (row-zero, equivalently p_T as a matrix coefficient annihilated from the dual side) reflects the left-multiplicative absorption that “Diagnostic 3” formalises. Both are clean Hecke-algebra facts. The 108 zero diagonals decompose as $|\ker \Omega| + |\ker \Omega^*| - |\ker \Omega \cap \ker \Omega^*| = 105 + 48 - 45 = 108$ on the seminormal basis. The two “interior D-class” zeros that prior work flagged as structurally unexplained, $T = ((1, 2, 3), (4, 5, 8), (6), (7), (9))$ and $T = ((1, 2, 3), (4, 6, 8), (5), (7), (9))$, are exactly the two basis vectors in $\ker \Omega^* \setminus \ker \Omega$. They are not killed by the column mechanism; they *are* killed by the row mechanism. We give a proof that the row-zero side is a sufficient condition (the trivial direction of the dichotomy), exhibit the algebraic structure $B := \Omega V^\lambda$ has rank 2 (so $\dim B = 2$) which is what makes the kernel structure so clean here, and leave the combinatorial characterisation of $\ker \Omega^*$ in the Pillar-1 regime open as Conjecture 1. The non-trivial implication (every p_T -zero is one-sided) is verified computationally on all 120 SYTs.

1 Setup and the main theorem

We follow the conventions of [3, 1, 4].

1.1 Hecke algebra and the chain operator

$H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$, braid and commutation relations, and quadratic $T_i^2 = (q-1)T_i + q$. We set $S_i := T_i + 1$; equivalently,

$$S_i^2 = (q+1)S_i, \quad T_i S_i = q S_i = S_i T_i, \tag{1}$$

so $\text{im}(S_i) = \ker(T_i - q) =: E_i^+$ and $\ker(S_i) = \ker(T_i + 1) =: E_i^-$. The Specht module V^λ decomposes $V^\lambda = E_i^+ \oplus E_i^-$.

For a SYT T of shape λ , v_T denotes the Hoefsmit ($b' = 1$) seminormal basis vector. The local action of S_i on v_T depends only on the relative positions of $i, i+1$ in T : **same row** (SR) gives $S_i v_T = (q+1)v_T \in E_i^+$; **same column** (SC) gives $S_i v_T = 0$, so $v_T \in E_i^-$; otherwise (**diagonal D**), S_i acts on the 2-dim block $\text{span}(v_T, v_{s_i T})$ as a non-degenerate matrix.

For $k \geq 2$, $R'_k := S_{k-1}S_{k-2} \cdots S_1$ and $L'_k := S_1S_2 \cdots S_{k-1}$, with $R'_1 = L'_1 = 1$. Note that L'_k is the image of R'_k under the anti-involution $*$ defined by $(T_i)^* = T_i$ and $(ab)^* = b^*a^*$.

The chain operator on V^λ is

$$\Omega := R'_{\ell+1}R'_{\ell+2} \cdots R'_n, \quad \ell = \ell(\lambda) = \ell(\widehat{\lambda}) + q. \quad (2)$$

The anti-involution image is

$$\Omega^* = L'_n L'_{n-1} \cdots L'_{\ell+1}. \quad (3)$$

We write $p_T(q) := \langle v_T, \Omega v_T \rangle$ for the (T, T) -diagonal coefficient (the seminormal pairing is $\langle v_S, v_T \rangle = \delta_{ST}$).

1.2 The target shape

For $\lambda = (3, 3, 1, 1, 1)$ we have $n = 9$, $\ell = 5$, $\widehat{\lambda} = (3, 3)$, $|\widehat{\lambda}| = 6$, $\ell(\widehat{\lambda}) = 2$, $q = 3$ (trailing 1s), so $\tau(\widehat{\lambda}) = \max(0, 3 + 4 - 1 - 6) = 0$. The standing Pillar-1 hypothesis $\widehat{\lambda}_r \geq 2$ for all rows holds, and the q -bound $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2 = 4$ *fails* (we have $q = 3$). Pillar 1 is therefore not literally applicable, but the leftmost-factor SC theorem of [1] is unconditional, and the structural facts we use in this paper are all derived from the universal Hecke factorisations (2) and (3), requiring no τ -regime hypothesis.

1.3 Main theorem

Theorem 1 (Two-sided kernel dichotomy at $(3, 3, 1, 1, 1)$). *Let $\lambda = (3, 3, 1, 1, 1)$. For every $T \in \text{SYT}(\lambda)$,*

$$p_T(q) = 0 \iff v_T \in \ker(\Omega) \text{ or } v_T \in \ker(\Omega^*) \text{ in } V^\lambda. \quad (4)$$

On the 120-element seminormal basis, the kernels stratify as $|\{T : \Omega v_T = 0\}| = 105$, $|\{T : \Omega^ v_T = 0\}| = 48$, and $|\{T : \Omega v_T = 0 \wedge \Omega^* v_T = 0\}| = 45$, so by inclusion-exclusion the union has size $105 + 48 - 45 = 108$, matching the 108 zero diagonals exactly.*

In particular, the two “D-class interior zeros” of the survey,

$$\begin{aligned} T_a &= ((1, 2, 3), (4, 5, 8), (6), (7), (9)), \\ T_b &= ((1, 2, 3), (4, 6, 8), (5), (7), (9)), \end{aligned}$$

are exactly the elements of $\ker(\Omega^) \setminus \ker(\Omega)$ at this shape; Ωv_{T_a} and Ωv_{T_b} are nonzero in V^λ but the v_{T_a} - and v_{T_b} -coefficient functionals annihilate ΩV^λ .*

2 Proof of the sufficient direction

Lemma 2 (Column-zero). *If $\Omega v_T = 0$, then $p_T(q) = 0$.*

Proof. $p_T = \langle v_T, \Omega v_T \rangle = \langle v_T, 0 \rangle = 0$. □

Lemma 3 (Row-zero, trivial form). *If the row $\{\Omega_{T,S}\}_{S \in \text{SYT}(\lambda)}$ of the matrix of Ω in the seminormal basis is identically zero, then $p_T(q) = 0$.*

Proof. $p_T(q) = \Omega_{T,T}$ is one of the entries in row T . □

The non-trivial content is the connection to the anti-involution image Ω^* :

Proposition 4 (Row-zero $\Leftrightarrow$ anti-involution kernel). *Row T of Ω in the seminormal basis is identically zero if and only if $\Omega^*v_T = 0$ as a vector in V^λ .*

Proof. The Hoefsmit ($b' = 1$) seminormal basis and the symmetric ($b = 1$) seminormal basis differ by a diagonal rescaling $v_T^{(b'=1)} = d_T v_T^{(b=1)}$ for explicit nonzero scalars $d_T \in \mathbb{Q}(q)$. For any $X \in H_q$, the matrix entries transform as

$$X_{S,T}^{(b'=1)} = \frac{d_S}{d_T} X_{S,T}^{(b=1)}.$$

In particular, the zero pattern of every row and column of X is invariant under this rescaling.

In the symmetric basis, the bilinear pairing $\langle v_S, v_T \rangle_{(b=1)} = \delta_{ST}$ has the property that every T_i is self-adjoint ([6, §3]); equivalently the matrix of T_i in the symmetric basis is symmetric, hence equals its transpose. The transpose corresponds to the algebra anti-involution $*$, so $X_{S,T}^{*(b=1)} = X_{T,S}^{(b=1)}$. Therefore $\Omega^*v_T = 0$ in either basis iff column T of Ω^* in the ($b = 1$) matrix is zero iff row T of Ω in the ($b = 1$) matrix is zero iff (by zero-pattern invariance) row T of Ω in the ($b' = 1$) matrix is zero. $\square$ $\square$

Corollary 5 (Row-zero $\Rightarrow p_T = 0$). *If $\Omega^*v_T = 0$ in V^λ , then $p_T(q) = 0$.*

Proof. Combine Proposition 4 with Lemma 3. $\square$ $\square$

Corollary 5 packages “Diagnostic 3 SC plus the anti-involution mirror”: the SC case $\Omega^*v_T = 0$ via $S_\ell v_T = 0$ recovers [1, Thm. 1]; the new content is that the same mechanism applies to *any* $v_T \in \ker \Omega^*$, not just to SC tableaux.

The $\Leftarrow$ direction of Theorem 1 now follows by the union of Lemmas 2 and 5.

3 Structural origin of the two kernels

3.1 The column kernel

Proposition 6 (Sufficient column-zero conditions). *Let $T \in \text{SYT}(\lambda)$. If pair $(1, 2)$ is SC at T , then $\Omega v_T = 0$. More generally, if there exists $k \in \{2, \dots, n\}$ such that $R'_k v_T \in \ker(R'_{k+1} R'_{k+2} \cdots R'_n)$ (with the convention $R'_{n+1} \cdots R'_n = I$), then $\Omega v_T = 0$.*

Proof. $\Omega = R'_{\ell+1} \cdots R'_{n-1} \cdot R'_n$, and $R'_n = S_{n-1} \cdots S_2 S_1$ ends in S_1 . If pair $(1, 2)$ is SC at T , then $S_1 v_T = 0$, so $R'_n v_T = 0$, so $\Omega v_T = 0$. The general statement is the iterated version. $\square$ $\square$

3.2 The row kernel

Proposition 7 (Sufficient row-zero conditions). *Let $T \in \text{SYT}(\lambda)$. If pair $(\ell, \ell + 1)$ is SC at T (i.e., T is SC at $\ell = 5$ for our shape), then $\Omega^*v_T = 0$.*

Proof. $\Omega^* = L'_n L'_{n-1} \cdots L'_{\ell+1}$, and $L'_{\ell+1} = S_1 S_2 \cdots S_\ell$ ends (on the right) in S_ℓ . If $S_\ell v_T = 0$, then $L'_{\ell+1} v_T = 0$, hence $\Omega^*v_T = 0$. $\square$ $\square$

Proposition 7 recovers [1, Thm. 1] via Lemma 5: it is literally Diagnostic 3 SC at the ℓ -side, derived from the rightmost factor of Ω^* .

3.3 Why $\dim B = 2$ at $(3, 3, 1, 1, 1)$ makes the dichotomy clean

The chain image $B := \Omega V^\lambda$ has $\dim B = 2$ at $\lambda = (3, 3, 1, 1, 1)$, verified by computing $\text{rank}(\Omega)$ symbolically in $\mathbb{Q}(q)$ and numerically at $q = 7$. This is a remarkable compression of the 120-dimensional Specht module to a 2-dimensional image, and is peculiar to this shape and parameter (cf. the rank-1 regime of Pillar 1, in which $B = 0$).

For $w \in B$ expanded in seminormal basis, the v_T -coefficient functional $v_T^* : B \rightarrow \mathbb{Q}(q)$ either vanishes identically (so $p_T = 0$ regardless of α_T, β_T in $\Omega v_T = \alpha_T w_1 + \beta_T w_2$) or it doesn't. The constraint “vanishes identically on B ” is equivalent to row T of Ω being zero, which by Lemma 5 is equivalent to $\Omega^* v_T = 0$.

4 Computational verification

We computed Ω as a 120×120 matrix over $\mathbb{Q}(q)$ in the Hoefsmit seminormal basis. The non-kernel basis vectors (i.e., the indices T with $\Omega v_T \neq 0$) are the 15 indices

$$\{0, 1, 2, 4, 5, 10, 11, 12, 14, 15, 20, 21, 22, 24, 25\}$$

in the lexicographic SYT enumeration of [5, `syt_of_shape`]. Of these, the 12 indices $\{0, 1, 4, 10, 11, 12, 14, 15, 21, 22, 24, 25\}$ have nonzero $p_T(q)$, and the three indices $\{2, 5, 20\}$ have $p_T(q) = 0$. The zero-row set (basis vectors v_T with row T of Ω identically zero) has 48 elements; intersected with $\{0, \dots, 119\} \setminus \ker(\Omega)$ it yields exactly $\{2, 5, 20\}$.

In particular:

- Index 20 is the historical mystery tableau $T_0 = ((1, 2, 5), (3, 4, 6), (7), (8), (9))$, an SC tableau (pair $(5, 6)$ same-column); $\Omega^* v_{T_0} = 0$ via Proposition 7.
- Indices 2 and 5 are the D-class interior zeros T_a, T_b flagged in the abstract; for these, pair $(5, 6)$ is D (5 at $(1, 1)$ and 6 at $(2, 0)$, resp. 5 at $(2, 0)$ and 6 at $(1, 1)$), so Proposition 7 does *not* apply directly. The mechanism for $\Omega^* v_T = 0$ at $T = T_a, T_b$ is a deeper Hecke cancellation in $L'_n L'_{n-1} L'_{n-2} \cdot L'_{\ell+1} v_T$, left open as Conjecture 8 below.

Verification scripts: `~/projects/scratch/2026-05-18-d-class-zeros/` (`support_analysis.py`, `kernel_structure.py`, `check_anti_involution.py`, `check_anti_numeric.py`).

4.1 Distribution of D-class zeros

Of the 108 zero diagonals:

- 26 are SC at $(\ell, \ell + 1)$ (*closed* by Diagnostic 3 SC, [1]).
- 4 are SR at $(\ell, \ell + 1)$ (*closed* by column-zero via pair- $(1, 2)$ SC, since all 4 also satisfy that condition; combined with descent-1 mechanics).
- 78 are D at $(\ell, \ell + 1)$ (the “D-class”):
 - 76 have $\Omega v_T = 0$ (column-zero, of which 52 have pair $(1, 2)$ SC, and 24 have pair $(1, 2)$ SR or D but are killed by deeper iterated rightmost-factor cancellation in $R'_7 R'_8 R'_9$).
 - **2** have $\Omega v_T \neq 0$ but row T of Ω is zero: these are the “interior D-class zeros” T_a, T_b , closed by Lemma 5.

The two interior D-class zeros were the explicit open problem at the head of the May-17 SC paper’s discussion section [1, §5.3]; Theorem 1 together with Lemma 5 closes them.

Table 1: Cardinality breakdown of the two-sided dichotomy at $\lambda = (3, 3, 1, 1, 1)$ ($n = 9$, $\ell = 5$, $|\text{SYT}| = 120$).

Stratum	Count	Explanation
$\ker(\Omega)$ basis	105	col-zero (Lemma 2)
$\ker(\Omega^*)$ basis	48	row-zero (Lemma 5)
Intersection	45	both
Union	108	all zero p_T
Neither	12	all nonzero p_T
Of the 105 column-zero: pair (1, 2) SC	70	descent-1 trivial
... deeper col cancellation	35	
Of the 48 row-zero: pair (5, 6) SC	26	Prop. 7
... deeper row cancellation	22	
$\ker(\Omega^*) \setminus \ker(\Omega)$	3	interior zeros $\{2, 5, 20\}$
among these, D-class	2	T_a, T_b (today's targets)
among these, SC	1	T_0 (May 17 smoking gun)

5 Open: the structural characterisation of $\ker(\Omega^*)$

The sufficient direction of Theorem 1 is now proved (Section 2). The full equivalence is verified computationally at $\lambda = (3, 3, 1, 1, 1)$ but is not yet proved structurally. The remaining gap is a clean combinatorial description of $\ker(\Omega^*)$ on V^λ .

Conjecture 8 (Stratification of $\ker(\Omega^*)$). *For $\lambda = (\hat{\lambda}, 1^q)$ in the standing hypothesis, the kernel $\ker(\Omega^*) \subseteq V^\lambda$ stratifies according to “leftmost-factor elimination” in $L'_n \cdots L'_{\ell+1}$: a basis vector v_T satisfies $\Omega^* v_T = 0$ iff there exists $k \in \{\ell+1, \dots, n\}$ such that the partial product $L'_k L'_{k-1} \cdots L'_{\ell+1}$ applied to v_T lies in $\ker(L'_{k+1})$ in V^λ (with the convention $L'_{n+1} = I$).*

The $k = \ell + 1$ case is Proposition 7: $L'_{\ell+1} v_T = 0$. The $k > \ell + 1$ cases would explain the additional 22 row-zero basis vectors at $(3, 3, 1, 1, 1)$, including the T_a, T_b cases. A direct trace of $L'_n L'_{n-1} L'_{n-2} \cdots L'_{\ell+1} v_{T_a}$ should pin down the precise cancellation (a 2-block traversal through pair (5, 6) D, followed by a single S_4 killing the off-diagonal contribution at the $S_4 v_{s_5 T_a}$ step). We leave this as the strongest near-target generalisation.

Analogously, the column-zero side admits the conjecture:

Conjecture 9 (Stratification of $\ker(\Omega)$). *For $\lambda = (\hat{\lambda}, 1^q)$ in the standing hypothesis, $\ker(\Omega)$ stratifies by “rightmost-factor elimination” in $R'_{\ell+1} \cdots R'_n$ symmetrically: $\Omega v_T = 0$ iff there exists k with $R'_k v_T \in \ker(R'_{k+1} \cdots R'_{n-1} \cdot R'_n)$.*

The two conjectures together would give a fully combinatorial characterisation of zero diagonals, refining the survey diagnostics [4, §7] to the algorithmic “trace the cancellation chain on each side”.

Remark 10 ($\tau \geq 1$ generalisation). For $\hat{\lambda}$ with $\tau(\hat{\lambda}) \geq 1$ in the Pillar-1 regime, the rank $\dim B = 0$ (by the trace-vanishing forward theorem), so $B = 0$, every $p_T = 0$, and every basis vector is in $\ker(\Omega) \cap \ker(\Omega^*)$. The two-sided dichotomy is vacuously true. The interesting regime is $\tau = 0$ with $\dim B > 0$, where Theorem 1 provides a non-trivial decomposition.

Remark 11 (Why “row-zero” is the right mirror of Diagnostic 3). Diagnostic 3 in its survey form [4, §7] packages two facts: (a) the leftmost-factor SC theorem [1, Thm. 1] (“ $v_T \in E_\ell^- \Rightarrow p_T = 0$ ”), and (b) the SR companion [2, Thm. 1] (“ $v_T \in E_j^+$ for $j \leq \tau \Rightarrow p_T = 0$ ”). Both exploit *containment of one side* of $V^\lambda = E_i^+ \oplus E_i^-$ in either B (for (a)) or V^λ/B (for (b)). Lemma 5 is the precise mirror: instead of *one specific* eigenspace constraint, it allows *any* v_T annihilated by the full anti-involution image. The 26 SC tableaux are the subset of $\ker(\Omega^*)$ explained by the single-step S_ℓ -rightmost-factor; the additional 22 in $\ker(\Omega^*)$ at $(3, 3, 1, 1, 1)$ are the deeper cancellations that Conjecture 8 would explain.

References

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